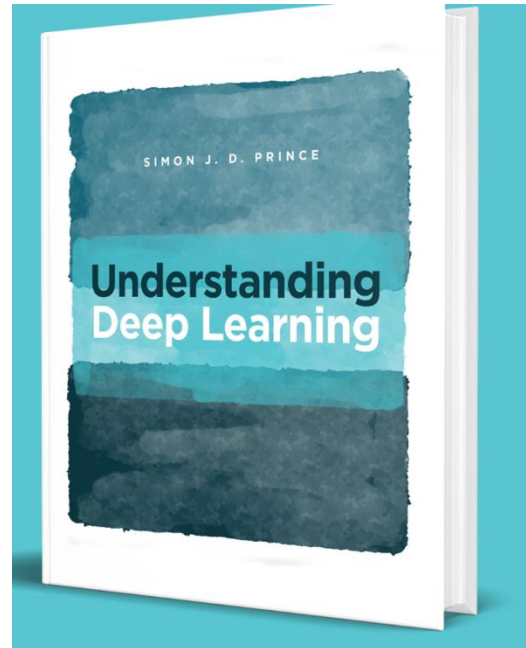


# Next slides are borrowed from



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@book{prince2023understanding,  
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  title = "Understanding Deep Learning",  
  publisher = "The MIT Press",  
  year = 2023,  
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CM20315 - Machine Learning  
Prof. Simon Prince



# Shallow neural networks

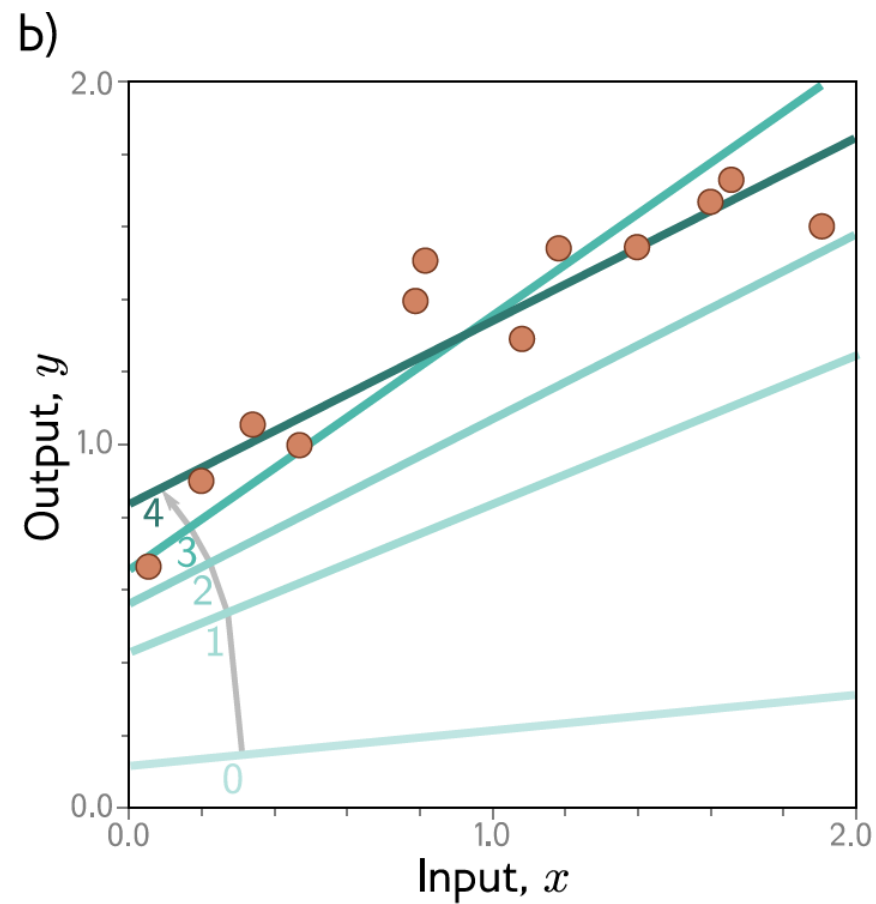
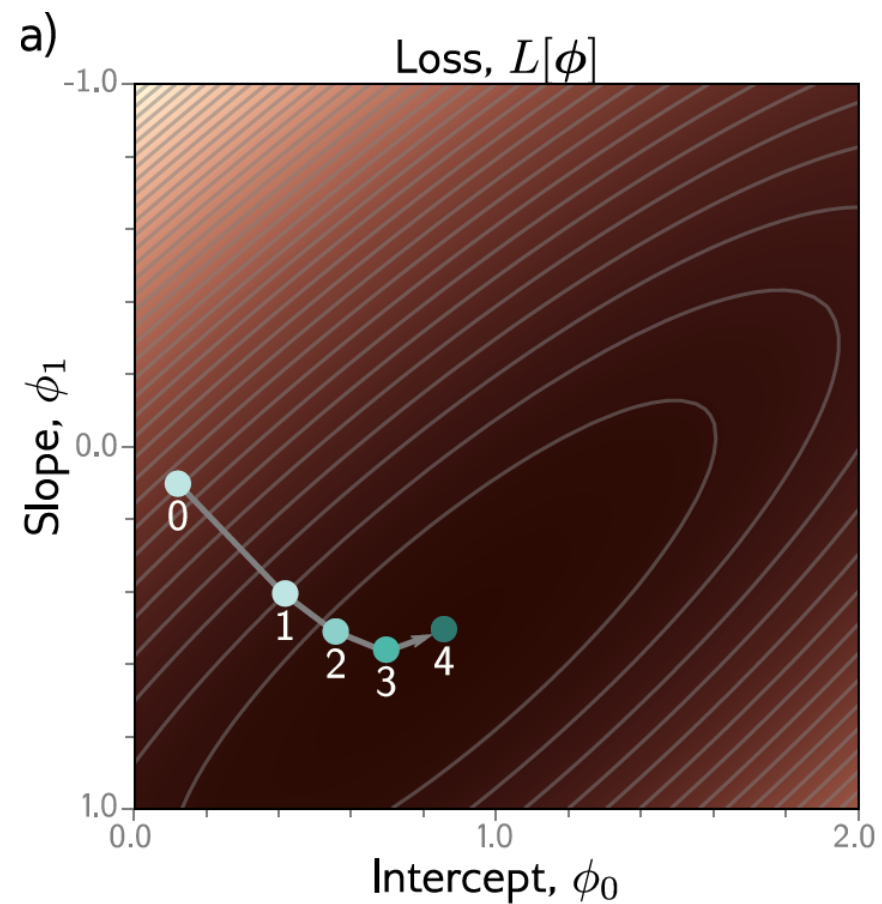
- 1D regression model is obviously limited
  - Want to be able to describe input/output that are not lines
  - Want multiple inputs
  - Want multiple outputs
- Shallow neural networks
  - Flexible enough to describe arbitrarily complex input/output mappings
  - Can have as many inputs as we want
  - Can have as many outputs as we want

# Shallow neural networks

- Example network, 1 input, 1 output
- Universal approximation theorem
- More than one output
- More than one input
- General case
- Number of regions
- Terminology

# 1D Linear Regression

$$y = f[x, \phi]$$
$$= \phi_0 + \phi_1 x$$



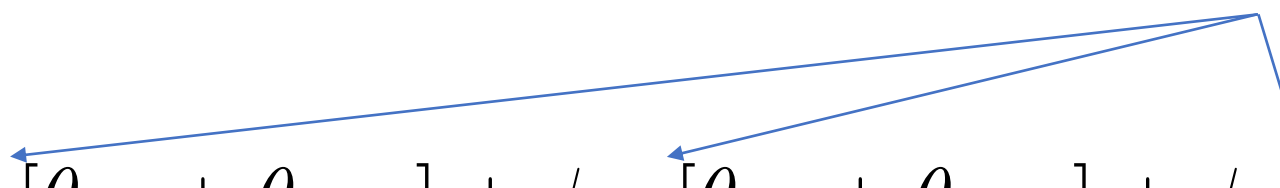
# Example shallow network

$$\begin{aligned} y &= f[x, \phi] \\ &= \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x] \end{aligned}$$

---

# Example shallow network

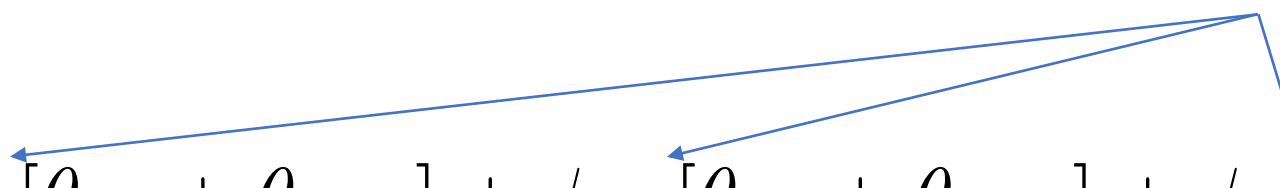
Activation function

$$\begin{aligned} y &= f[x, \phi] \\ &= \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x] \end{aligned}$$


The diagram consists of three blue arrows originating from the text 'Activation function' and pointing to the 'a' functions in the equation:  $a[\theta_{10} + \theta_{11}x]$ ,  $a[\theta_{20} + \theta_{21}x]$ , and  $a[\theta_{30} + \theta_{31}x]$ . A horizontal blue line is positioned below the equation.

# Example shallow network

Activation function

$$\begin{aligned} y &= f[x, \phi] \\ &= \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x] \end{aligned}$$


---

$$a[z] = \text{ReLU}[z] = \begin{cases} 0 & z < 0 \\ z & z \geq 0 \end{cases}.$$

Rectified Linear Unit

(particular kind of activation function)

# Example shallow network

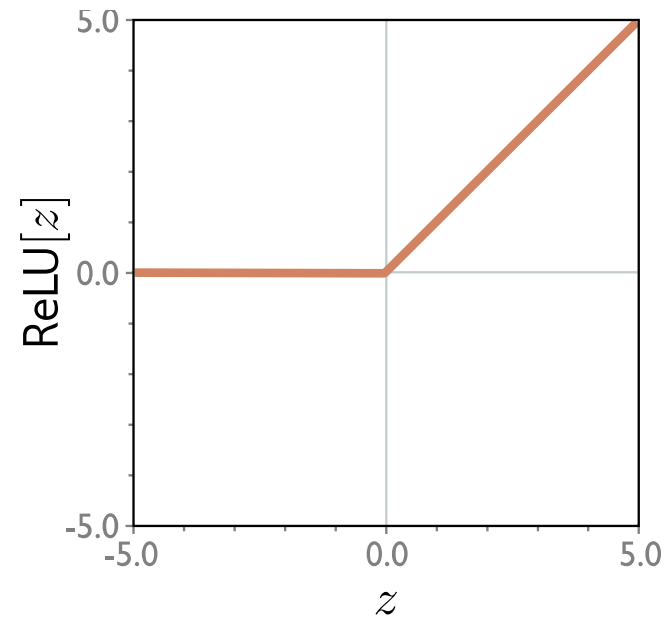
$$\begin{aligned} y &= f[x, \phi] \\ &= \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x] \end{aligned}$$

Activation function

$$a[z] = \text{ReLU}[z] = \begin{cases} 0 & z < 0 \\ z & z \geq 0 \end{cases}.$$

Rectified Linear Unit

(particular kind of activation function)



# Example shallow network

$$\begin{aligned} y &= f[x, \phi] \\ &= \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x] \end{aligned}$$

---

This model has 10 parameters:

$$\phi = \{\phi_0, \phi_1, \phi_2, \phi_3, \theta_{10}, \theta_{11}, \theta_{20}, \theta_{21}, \theta_{30}, \theta_{31}\}$$

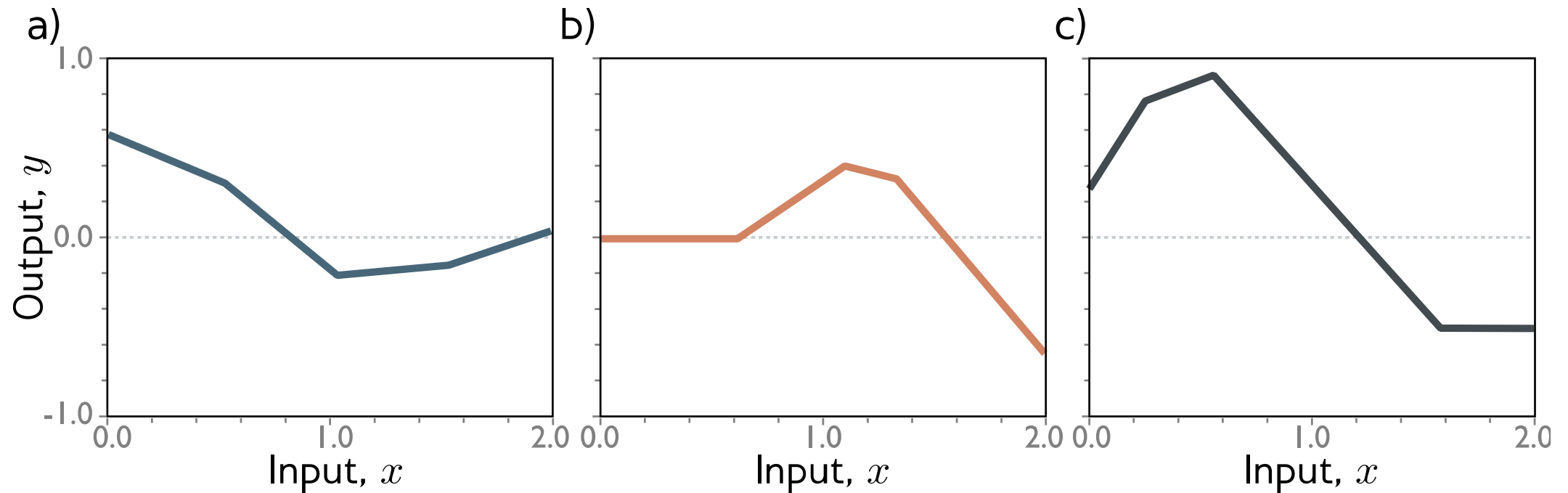
- Represents a family of functions
- Parameters determine particular function
- Given parameters can perform inference (run equation)
- Given training dataset  $\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^I$
- Define loss function  $L[\phi]$  (least squares)
- Change parameters to minimize loss function

# Example shallow network

$$y = \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x].$$

# Example shallow network

$$y = \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x].$$



Piecewise linear functions with three joints

# Hidden units

$$y = \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x].$$

---

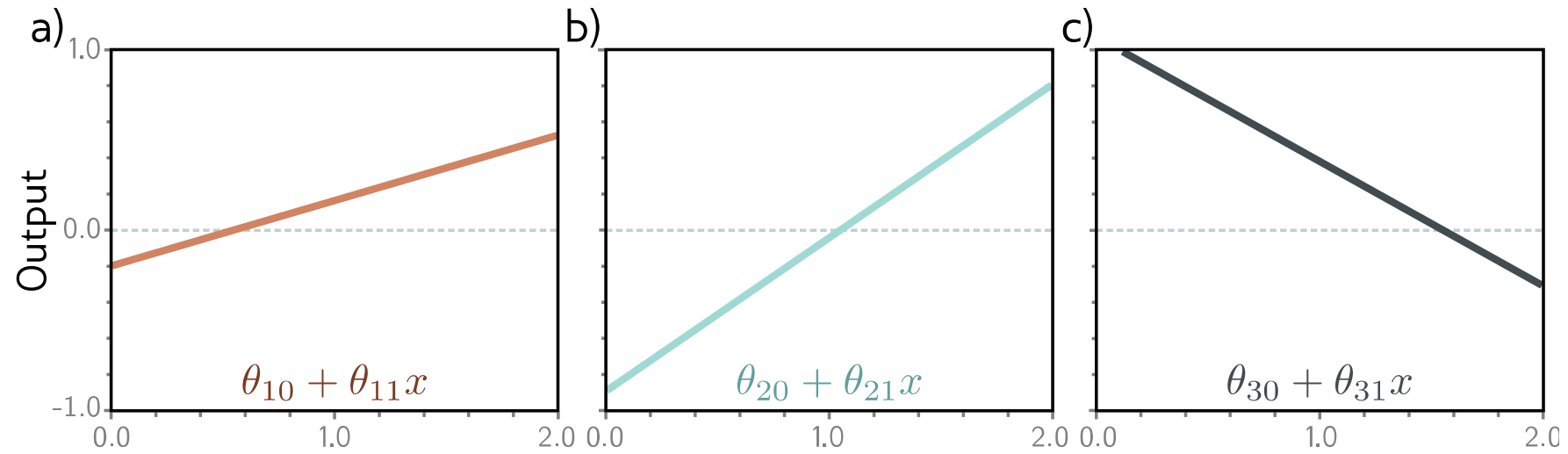
Break down into two parts:

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$

where:

$$\text{Hidden units} \left\{ \begin{array}{l} h_1 = a[\theta_{10} + \theta_{11}x] \\ h_2 = a[\theta_{20} + \theta_{21}x] \\ h_3 = a[\theta_{30} + \theta_{31}x] \end{array} \right.$$

1. compute three  
linear functions

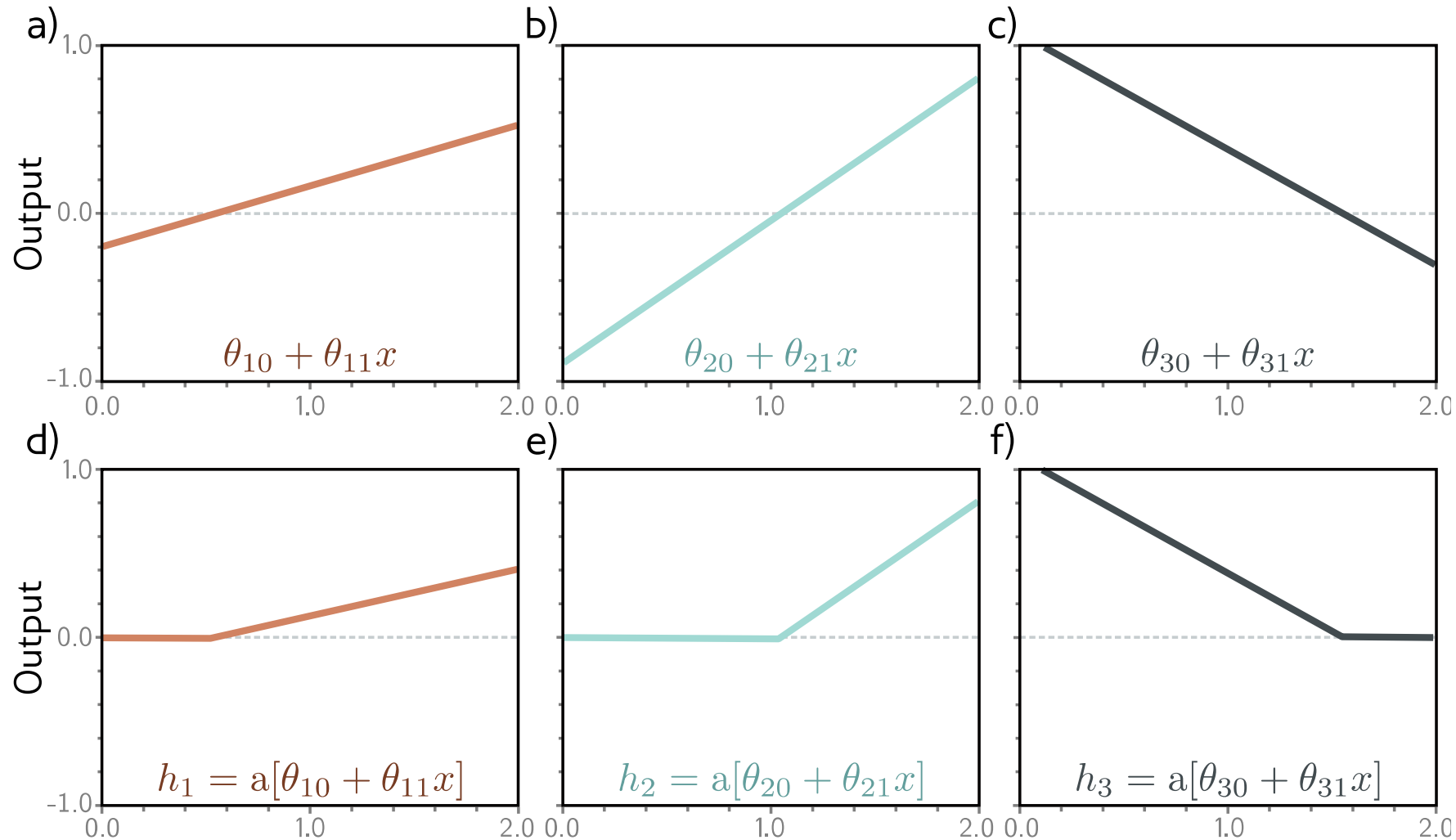


2. Pass through ReLU  
functions (creates  
hidden units)

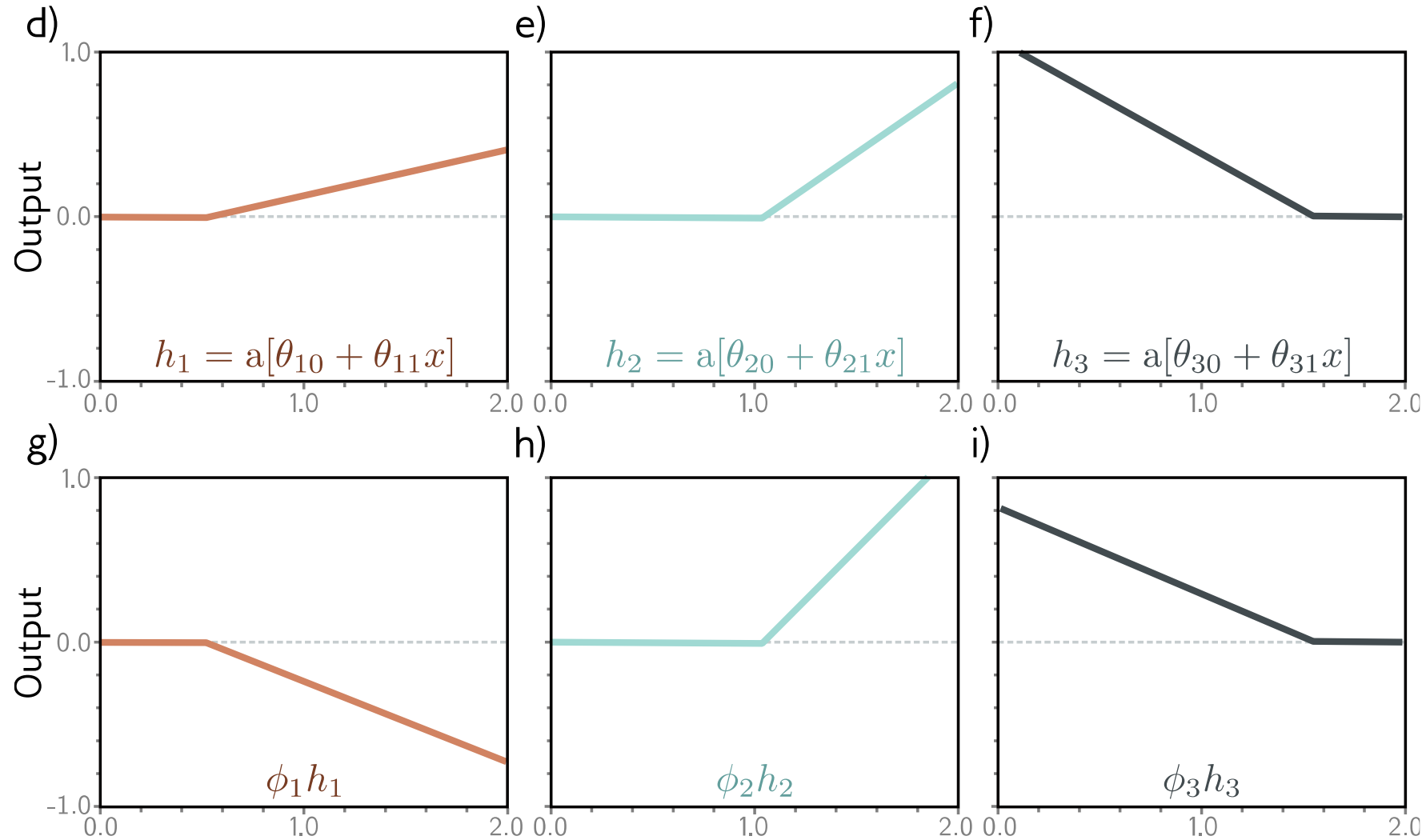
$$h_1 = a[\theta_{10} + \theta_{11}x]$$

$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x],$$

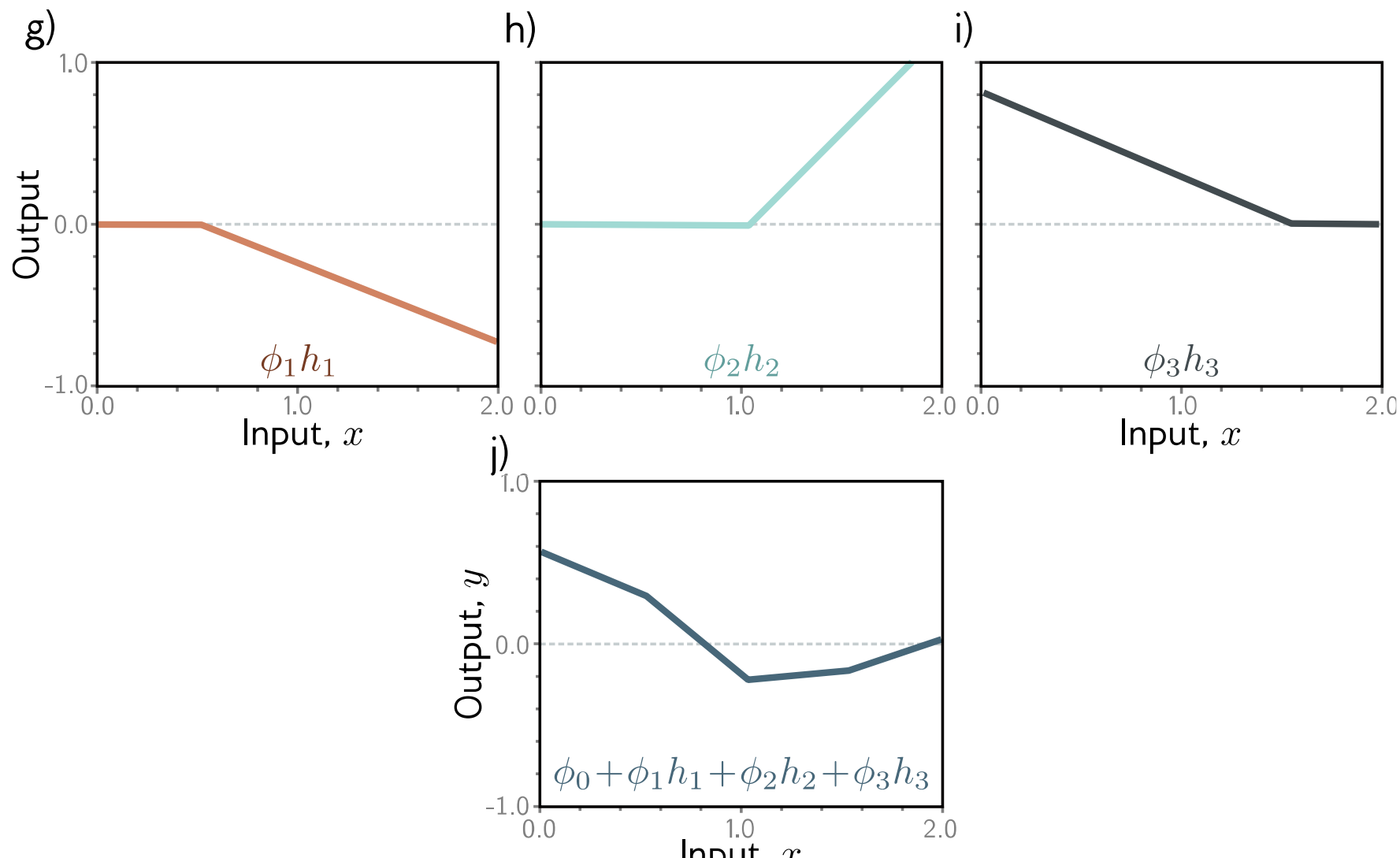


## 2. Weight the hidden units



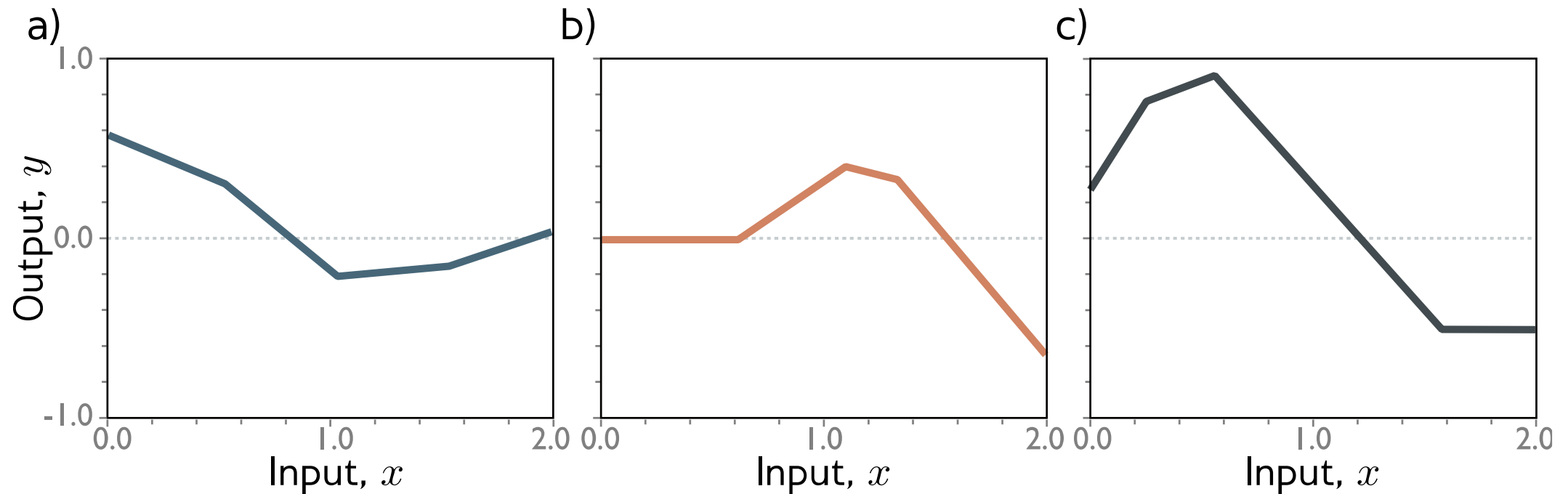
4. Sum the weighted  
hidden units to create  
output

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$



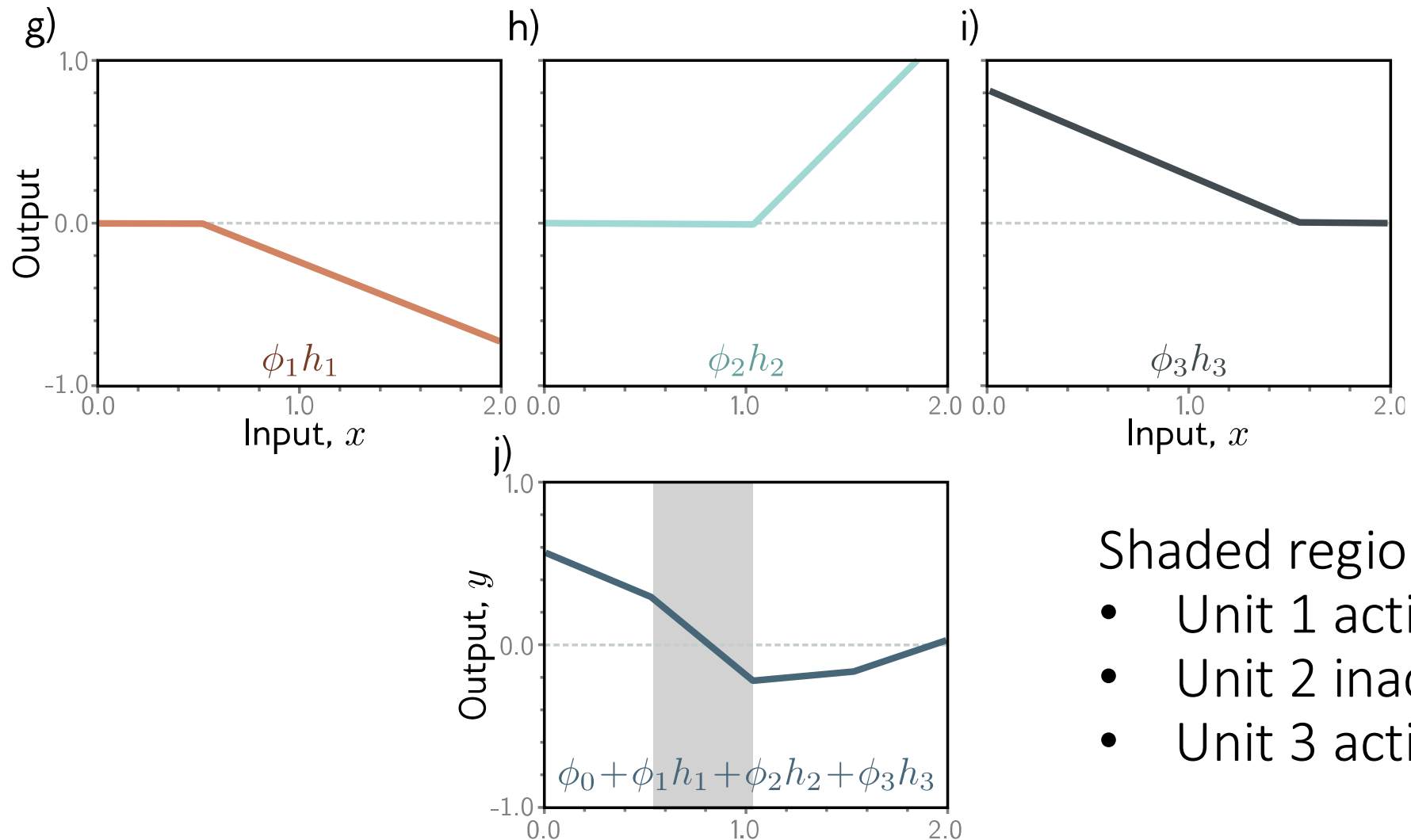
# Example shallow network

$$y = \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x].$$



Example shallow network = piecewise linear functions  
1 “joint” per ReLU function

Activation pattern = which hidden units are activated



Shaded region:

- Unit 1 active
- Unit 2 inactive
- Unit 3 active

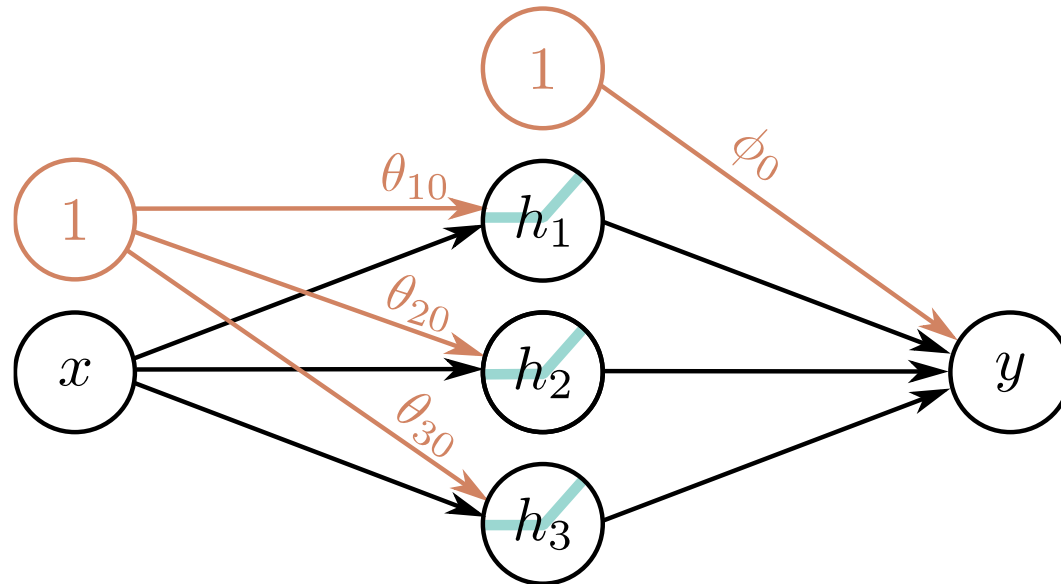
# Depicting neural networks

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$



Each parameter multiplies its source and adds to its target

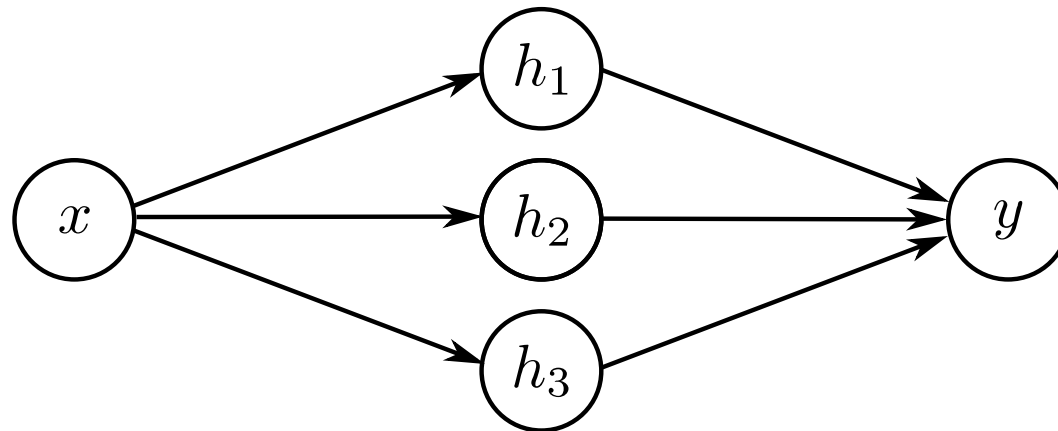
# Depicting neural networks

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$



# Shallow neural networks

- Example network, 1 input, 1 output
- Universal approximation theorem
- More than one output
- More than one input
- General case
- Number of regions
- Terminology

With 3 hidden units:

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$

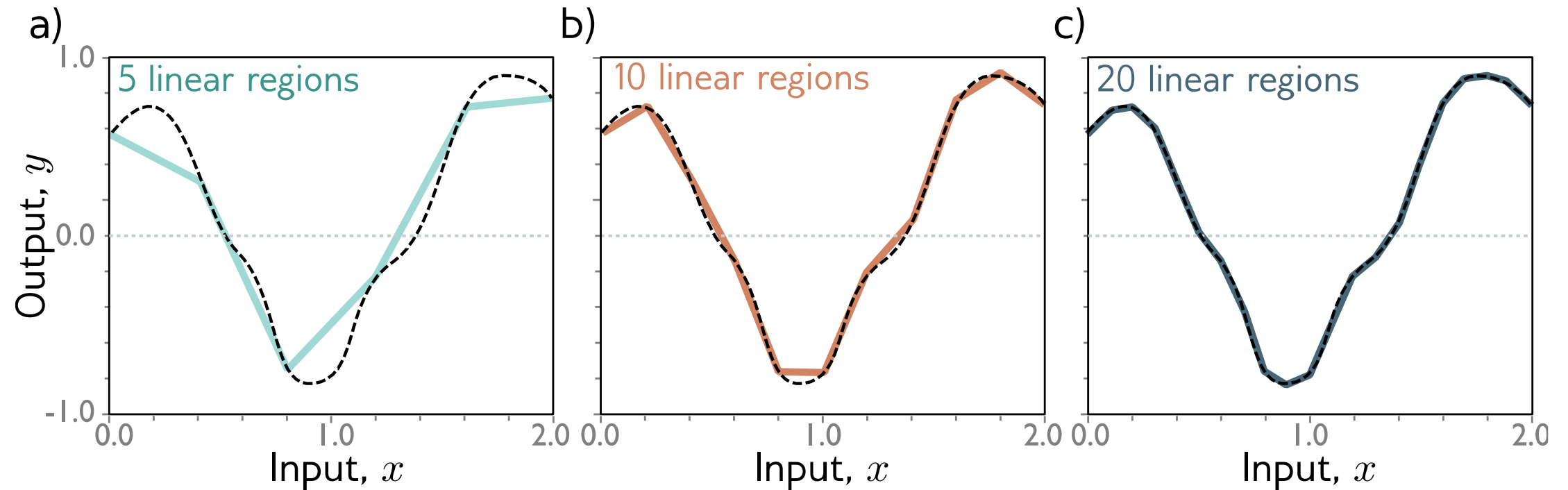
With D hidden units:

$$h_d = a[\theta_{d0} + \theta_{d1}x]$$

$$y = \phi_0 + \sum_{d=1}^D \phi_d h_d$$

# With enough hidden units...

... we can describe any 1D function to arbitrary accuracy



# Universal approximation theorem

“a formal proof that, with enough hidden units, a shallow neural network can describe any continuous function on a compact subset of  $\mathbb{R}^D$  to arbitrary precision”

# Shallow neural networks

- Example network, 1 input, 1 output
- Universal approximation theorem
- More than one output
- More than one input
- General case
- Number of regions
- Terminology

# Two outputs

- 1 input, 4 hidden units, 2 outputs

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$h_4 = a[\theta_{40} + \theta_{41}x]$$

$$y_1 = \phi_{10} + \phi_{11}h_1 + \phi_{12}h_2 + \phi_{13}h_3 + \phi_{14}h_4$$

$$y_2 = \phi_{20} + \phi_{21}h_1 + \phi_{22}h_2 + \phi_{23}h_3 + \phi_{24}h_4$$

# Two outputs

- 1 input, 4 hidden units, 2 outputs

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

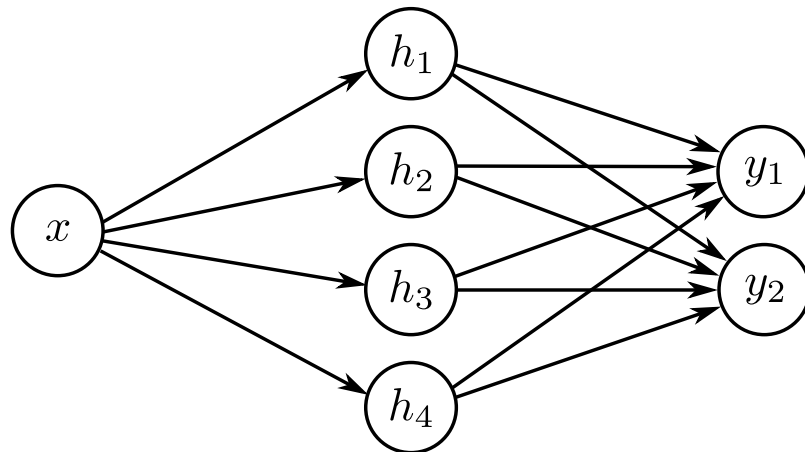
$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$h_4 = a[\theta_{40} + \theta_{41}x]$$

$$y_1 = \phi_{10} + \phi_{11}h_1 + \phi_{12}h_2 + \phi_{13}h_3 + \phi_{14}h_4$$

$$y_2 = \phi_{20} + \phi_{21}h_1 + \phi_{22}h_2 + \phi_{23}h_3 + \phi_{24}h_4$$



# Two outputs

- 1 input, 4 hidden units, 2 outputs

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

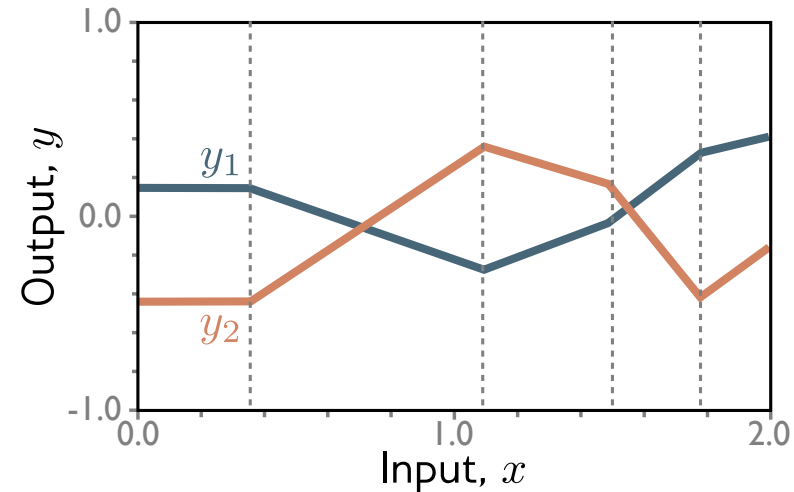
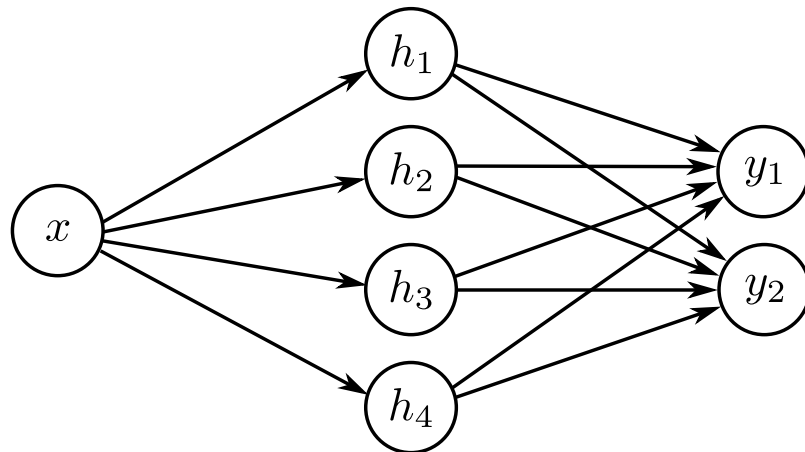
$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$h_4 = a[\theta_{40} + \theta_{41}x]$$

$$y_1 = \phi_{10} + \phi_{11}h_1 + \phi_{12}h_2 + \phi_{13}h_3 + \phi_{14}h_4$$

$$y_2 = \phi_{20} + \phi_{21}h_1 + \phi_{22}h_2 + \phi_{23}h_3 + \phi_{24}h_4$$



# Shallow neural networks

- Example network, 1 input, 1 output
- Universal approximation theorem
- More than one output
- More than one input
- General case
- Number of regions
- Terminology

# Two inputs

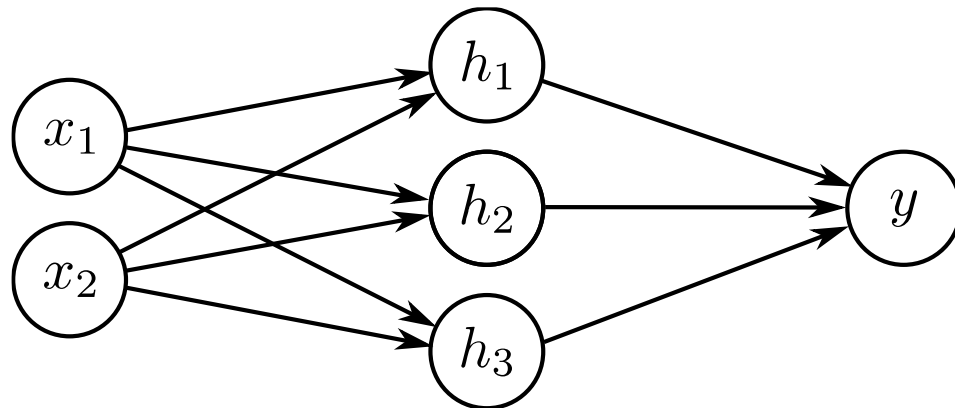
- 2 inputs, 3 hidden units, 1 output

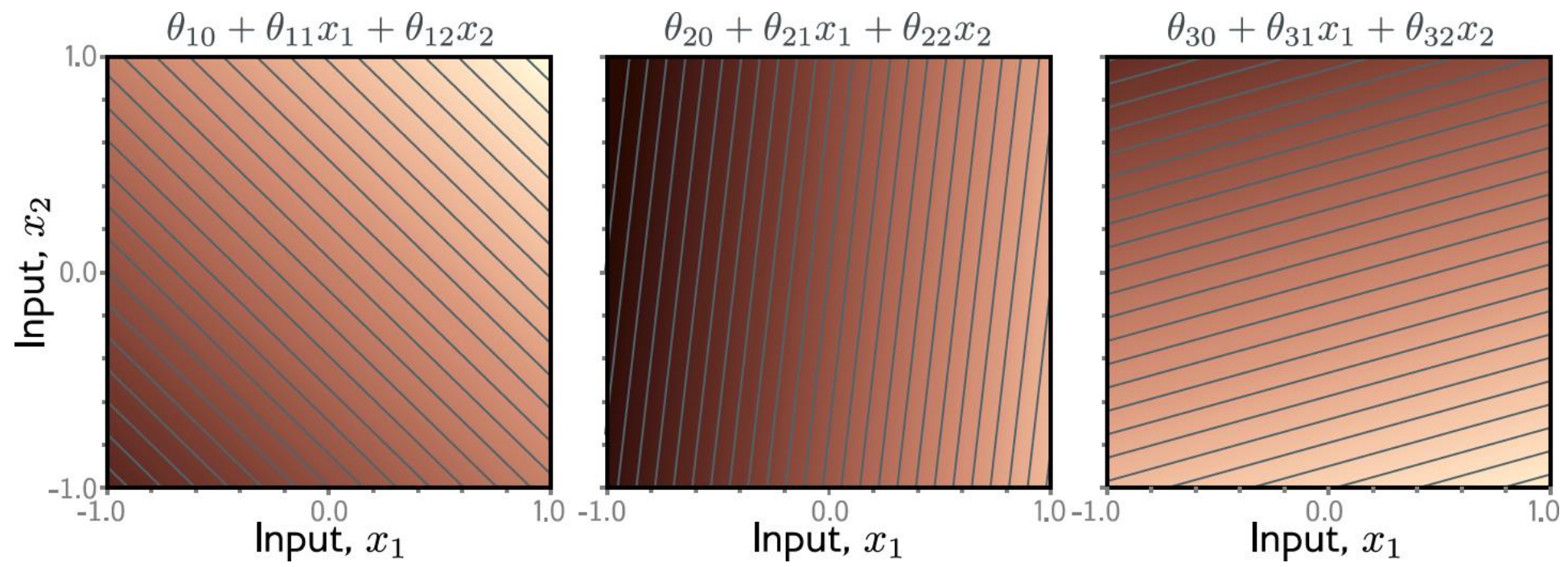
$$h_1 = a[\theta_{10} + \theta_{11}x_1 + \theta_{12}x_2]$$

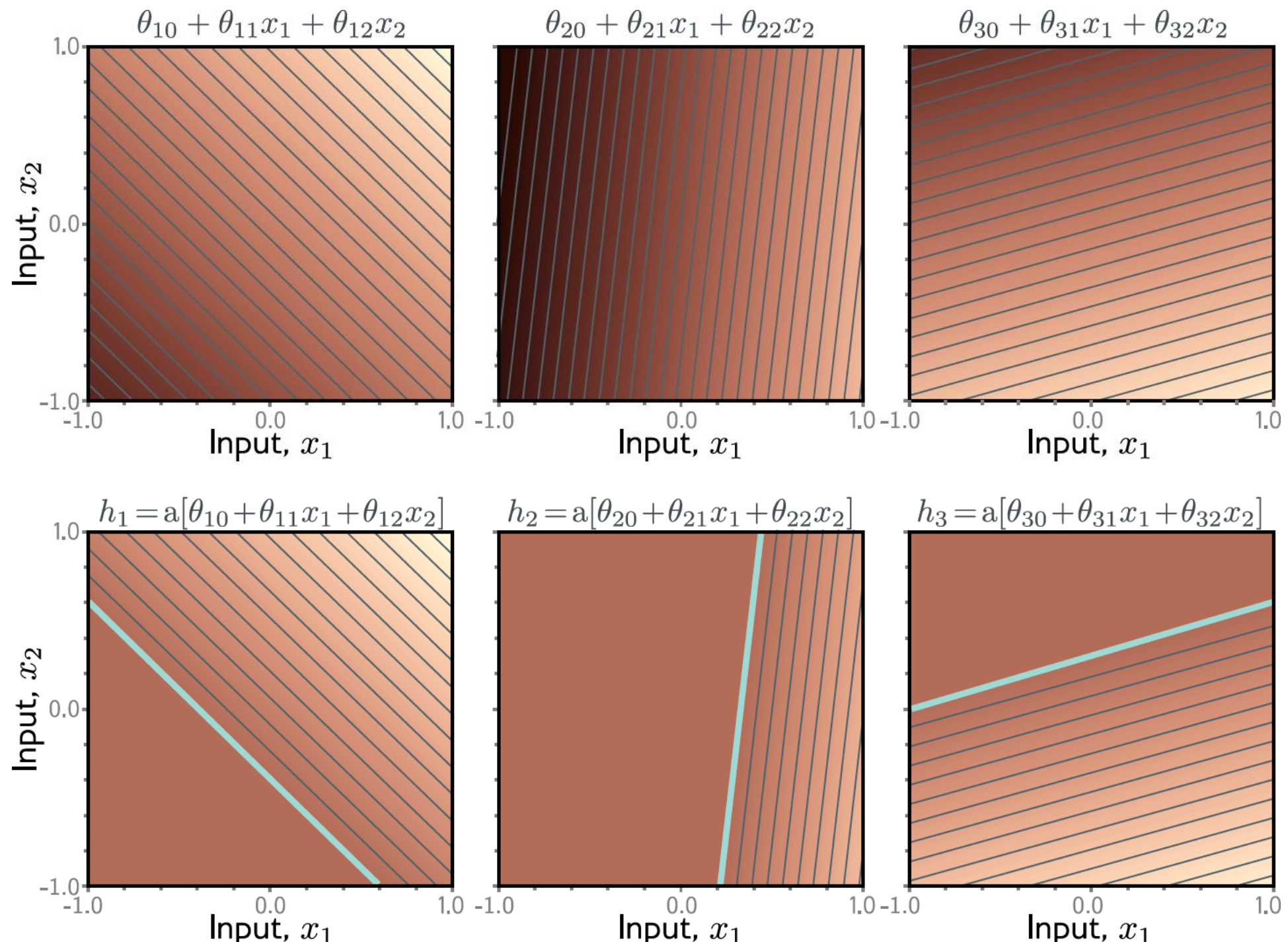
$$h_2 = a[\theta_{20} + \theta_{21}x_1 + \theta_{22}x_2]$$

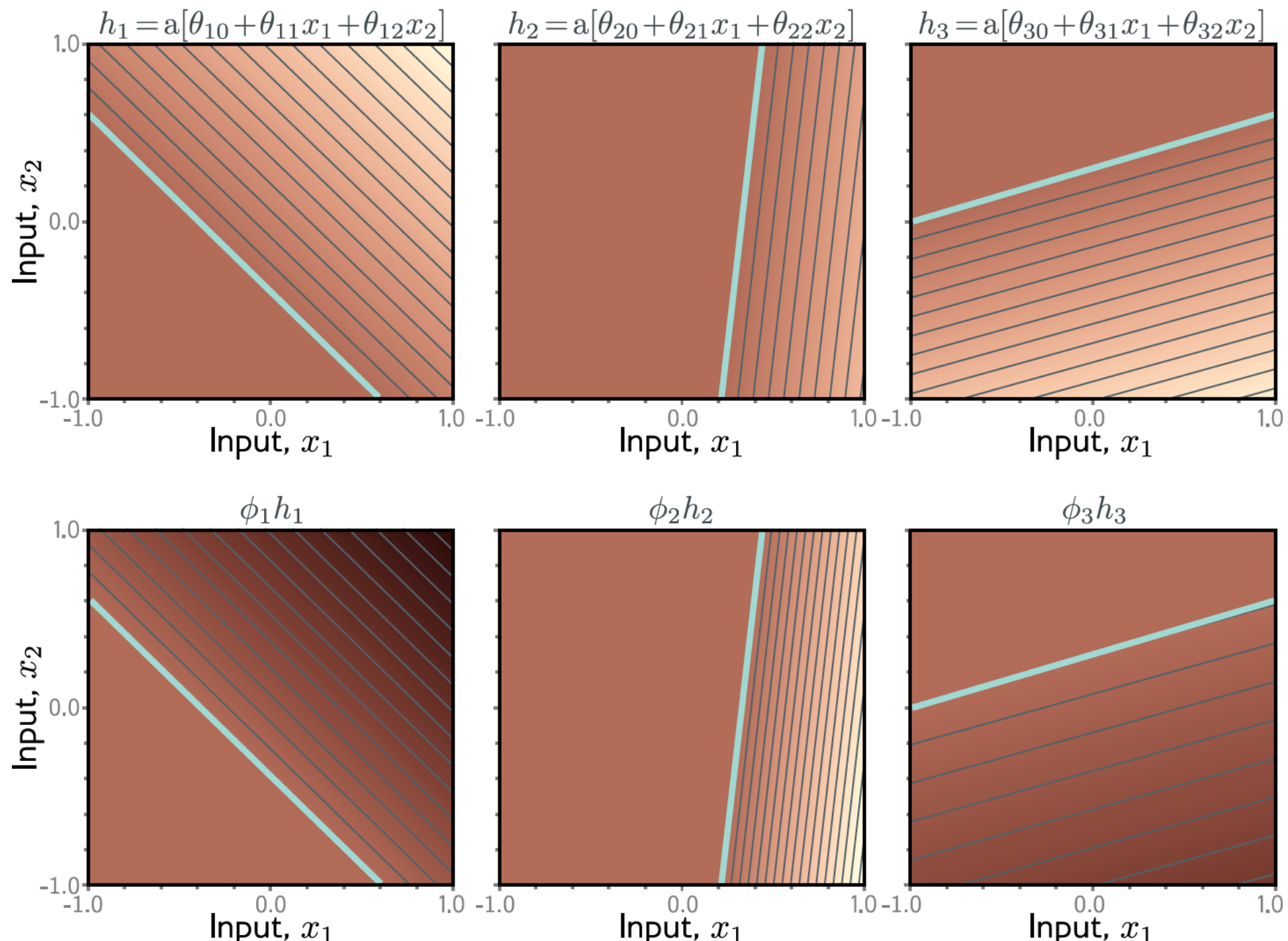
$$h_3 = a[\theta_{30} + \theta_{31}x_1 + \theta_{32}x_2]$$

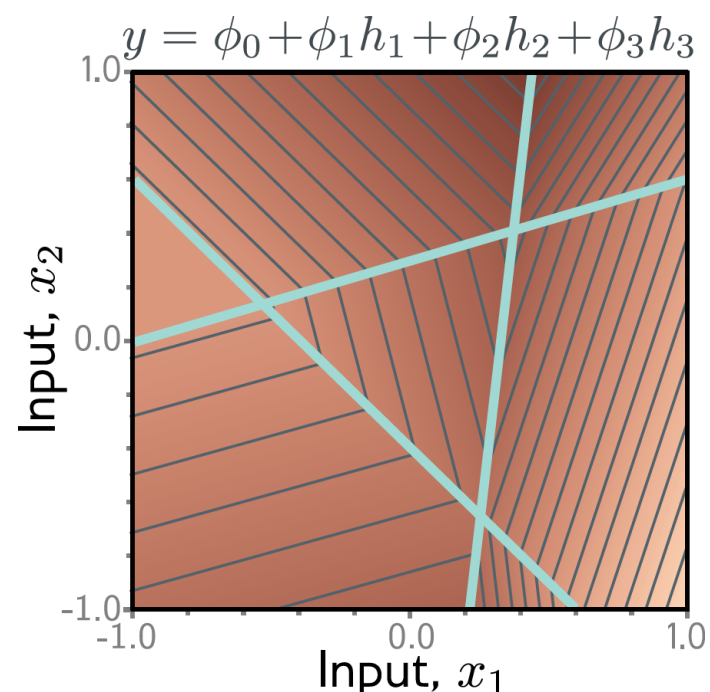
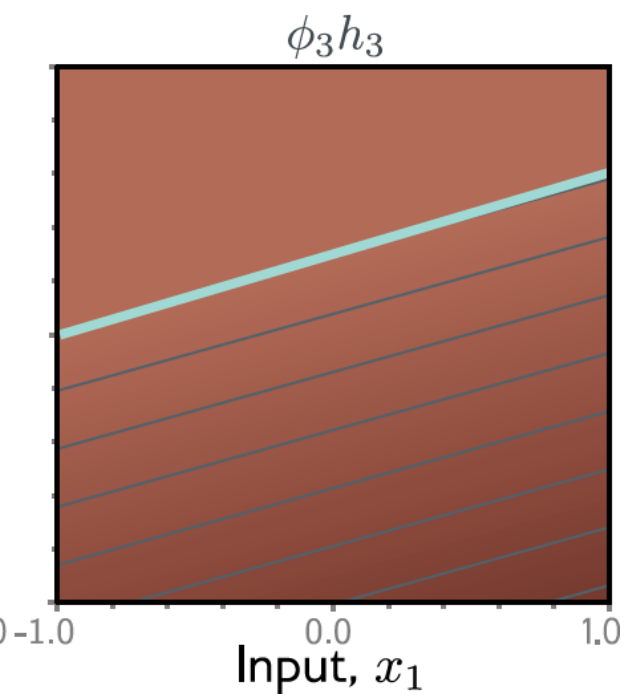
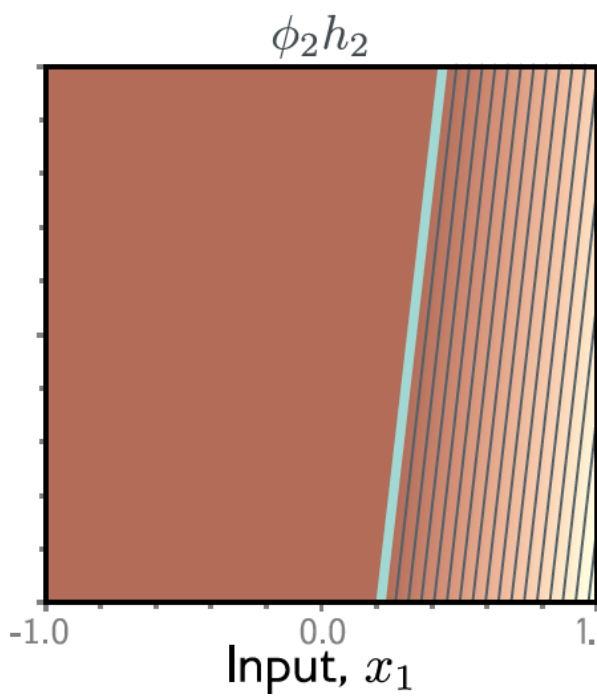
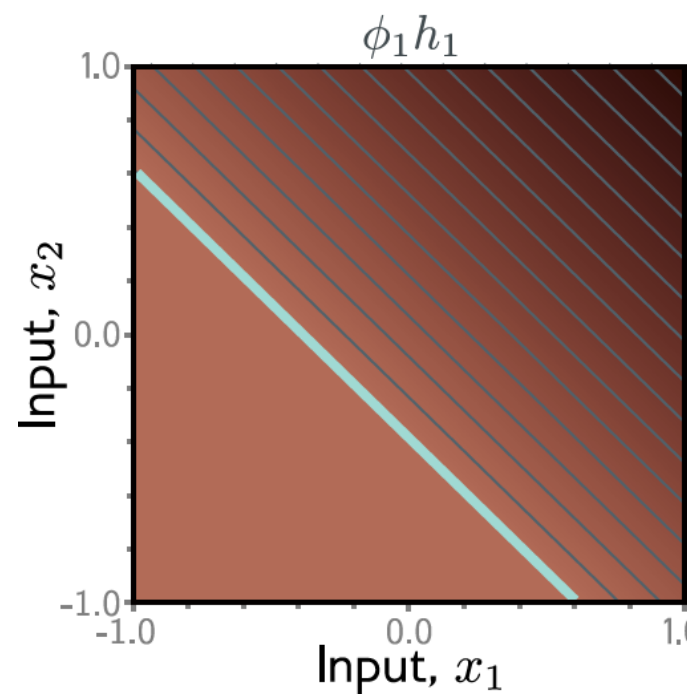
$$y = \phi_0 + \phi_1h_1 + \phi_2h_2 + \phi_3h_3$$

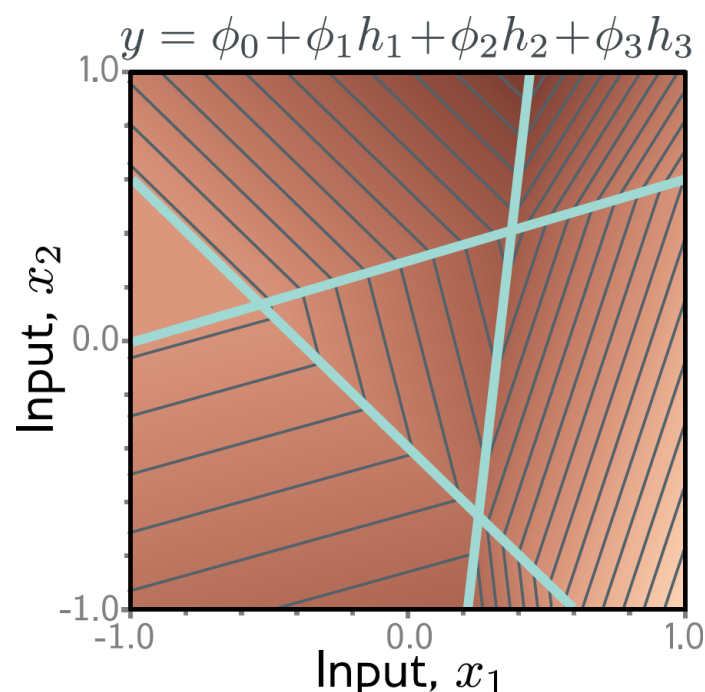
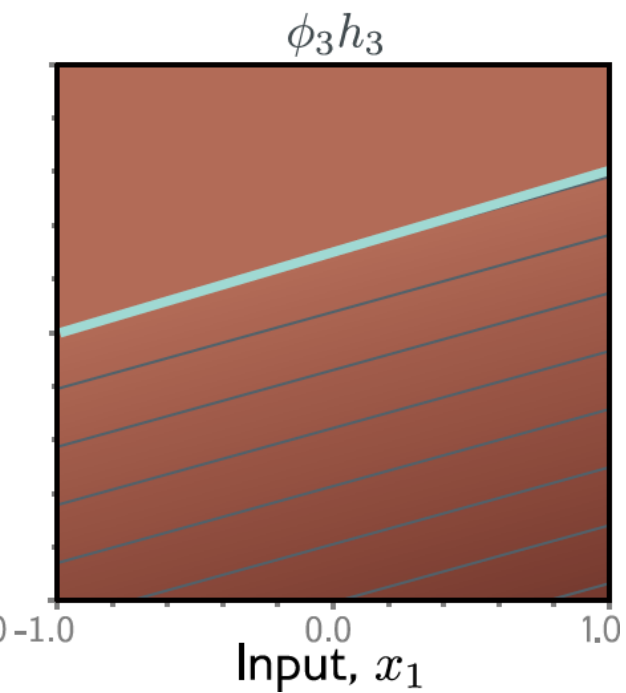
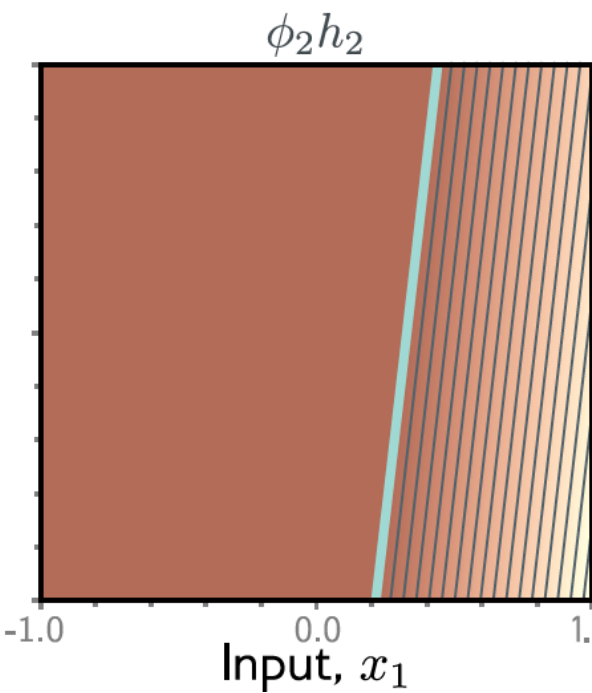
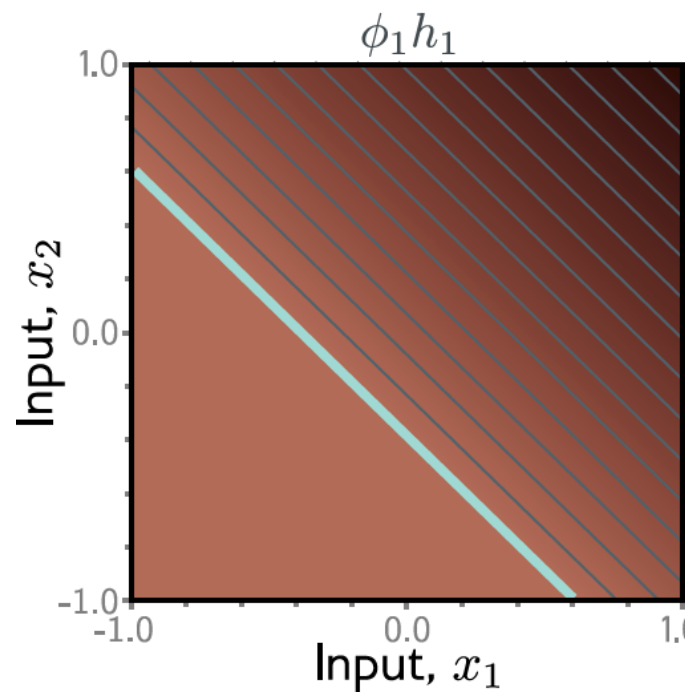






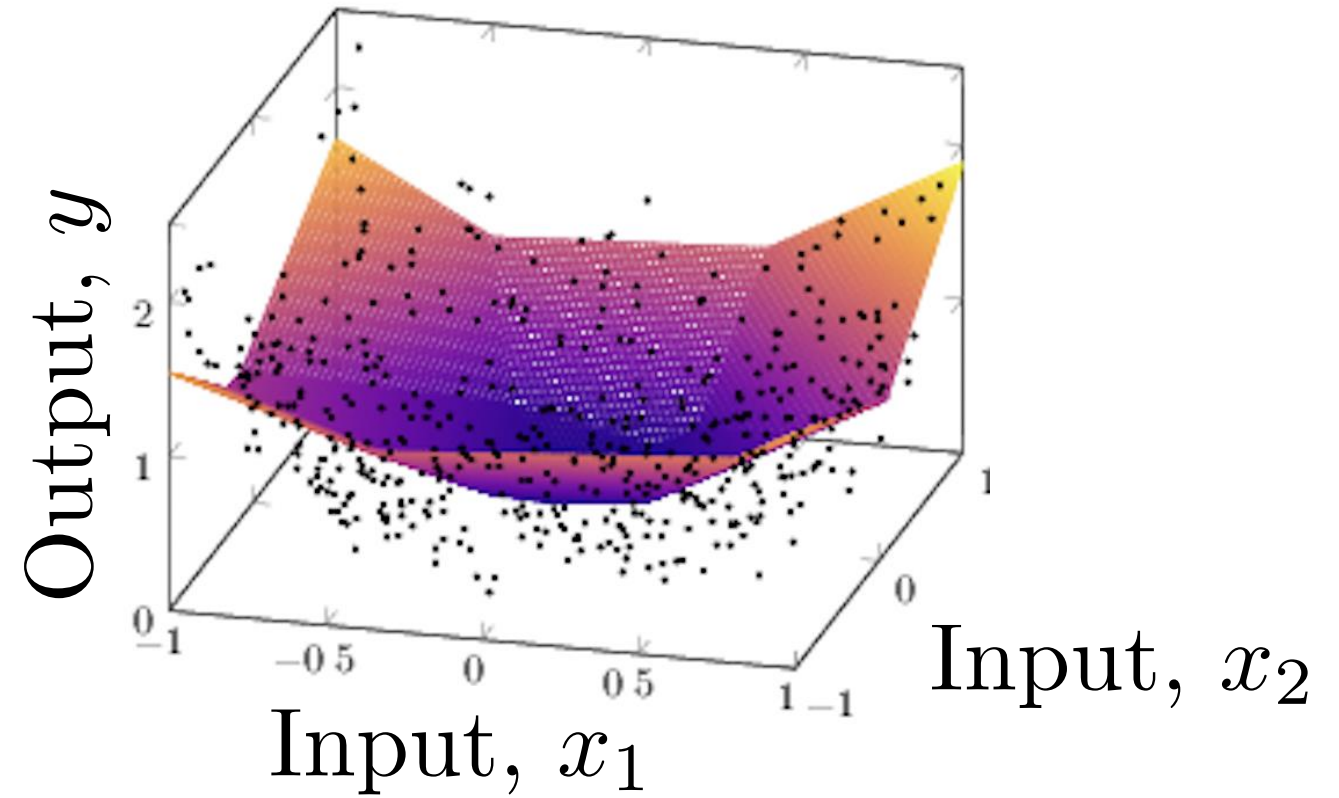
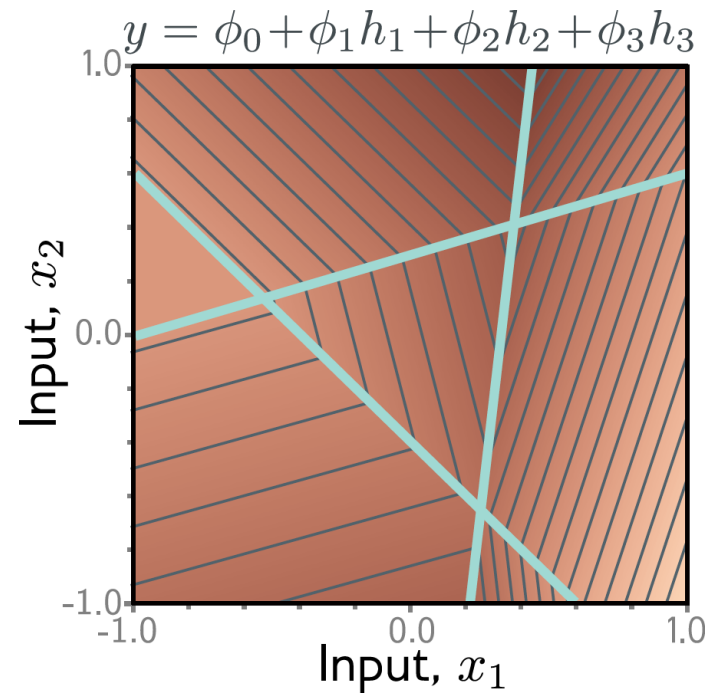






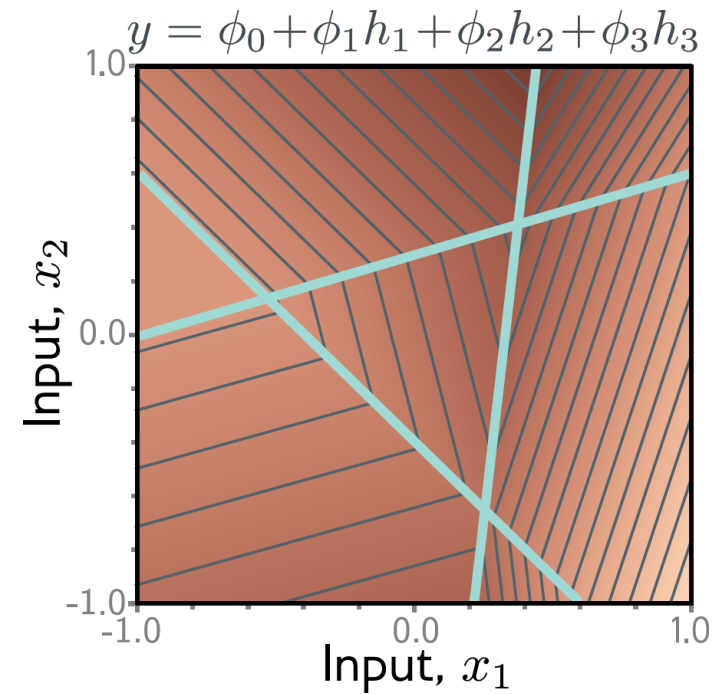
Convex polygons

# Fitting



# Question:

- For the 2D case, what if there were two outputs?
- If this is one of the outputs, what would the other one look like?



# Shallow neural networks

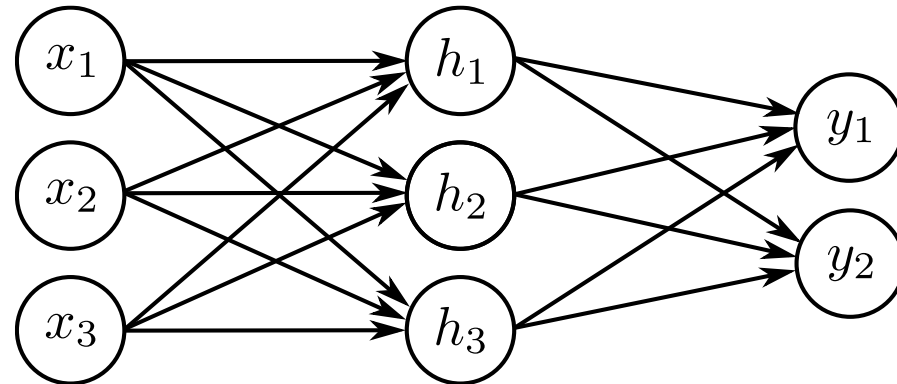
- Example network, 1 input, 1 output
- Universal approximation theorem
- More than one output
- More than one input
- General case
- Number of regions
- Terminology

# Arbitrary inputs, hidden units, outputs

- $D_o$  Outputs,  $D$  hidden units, and  $D_i$  inputs

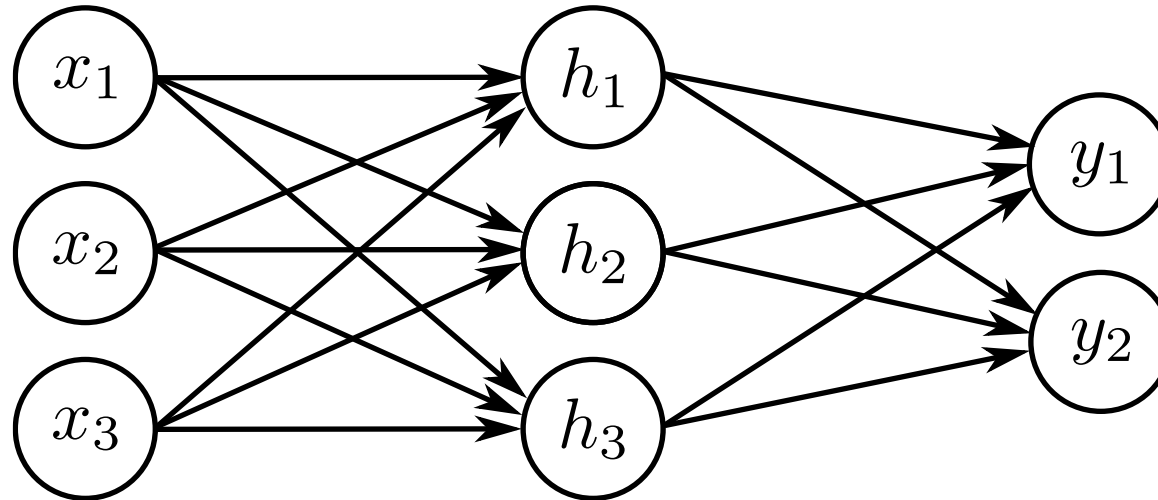
$$h_d = a \left[ \theta_{d0} + \sum_{i=1}^{D_i} \theta_{di} x_i \right] \quad y_j = \phi_{j0} + \sum_{d=1}^D \phi_{jd} h_d$$

- e.g., Three inputs, three hidden units, two outputs



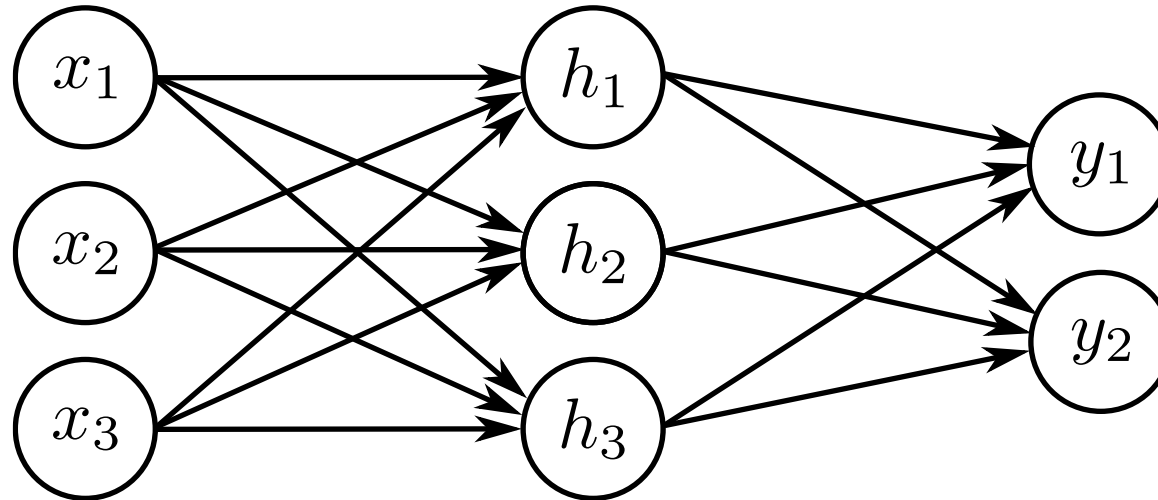
# Question 1:

- How many parameters does this model have?



## Question 2:

- Can you still visualize the functions / hidden units?

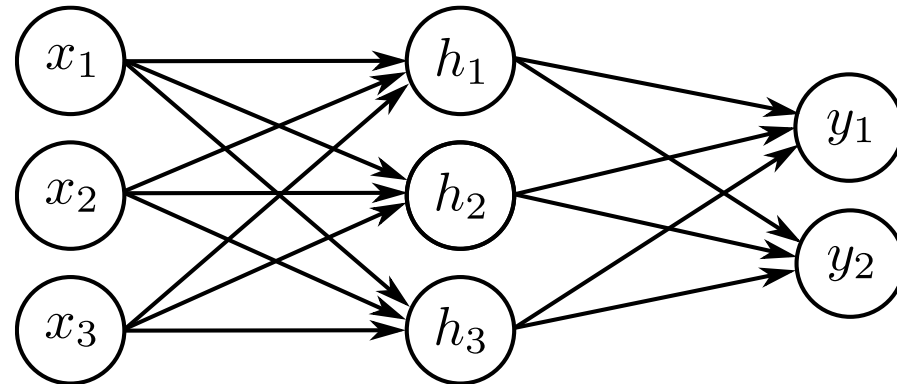


# Arbitrary inputs, hidden units, outputs

- $D_o$  Outputs,  $D$  hidden units, and  $D_i$  inputs

$$h_d = a \left[ \theta_{d0} + \sum_{i=1}^{D_i} \theta_{di} x_i \right] \quad y_j = \phi_{j0} + \sum_{d=1}^D \phi_{jd} h_d$$

- e.g., Three inputs, three hidden units, two outputs

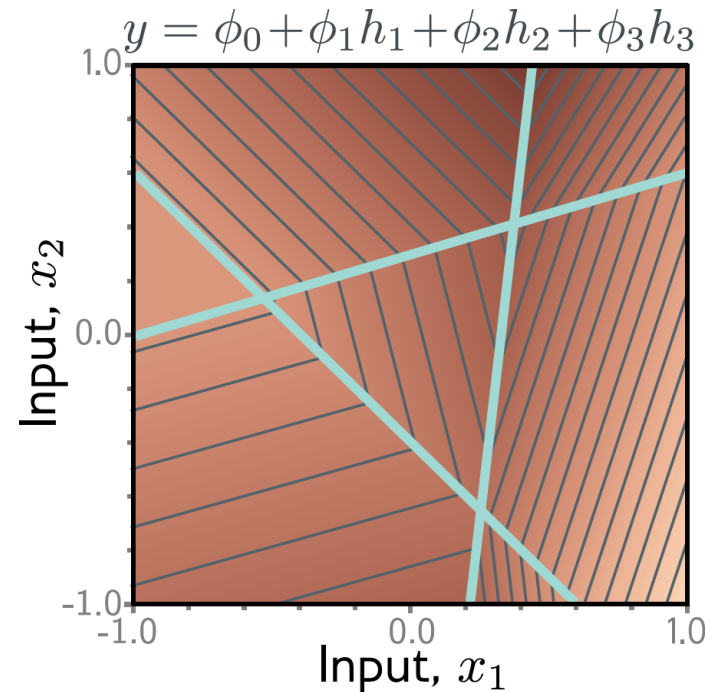


# Shallow neural networks

- Example network, 1 input, 1 output
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- More than one input
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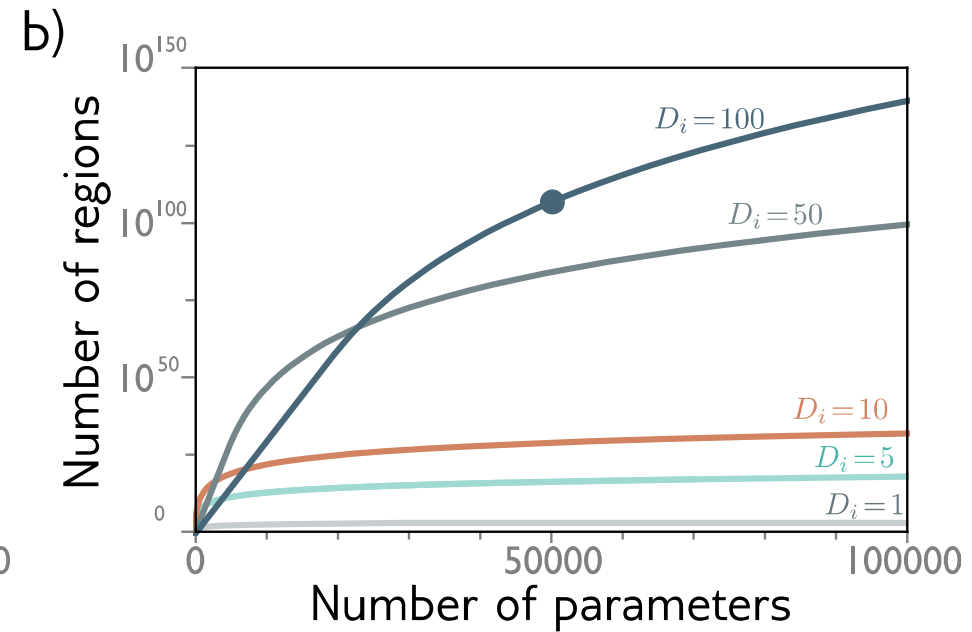
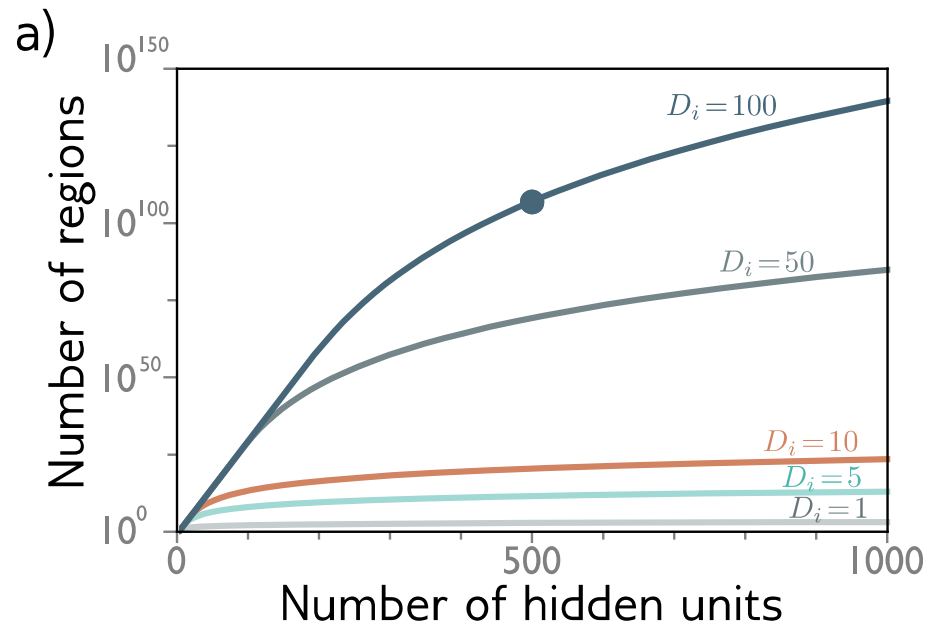
# Number of output regions

- In general, each output consists of D dimensional **convex polytopes**
- With two inputs, and three outputs, we saw there were seven polygons:



# Number of output regions

- In general, each output consists of  $D$  dimensional **convex polytopes**
- How many?



Highlighted point = 500 hidden units or 51,001 parameters

# Number of regions:

- Number of regions created by  $D > D_i$  planes in  $D_i$  dimensions was proved by Zaslavsky (1975) to be:

$$\sum_{j=0}^{D_i} \binom{D}{j} \leftarrow \text{Binomial coefficients!}$$

- How big is this? It's greater than  $2^{D_i}$  but less than  $2^D$ .

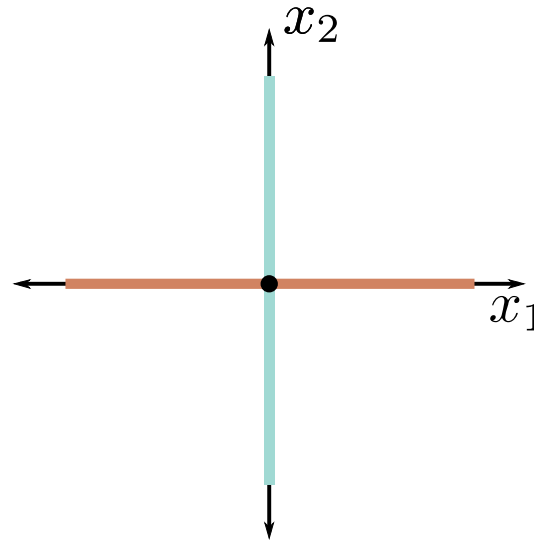
# Proof that more regions than $2^{Di}$

a)



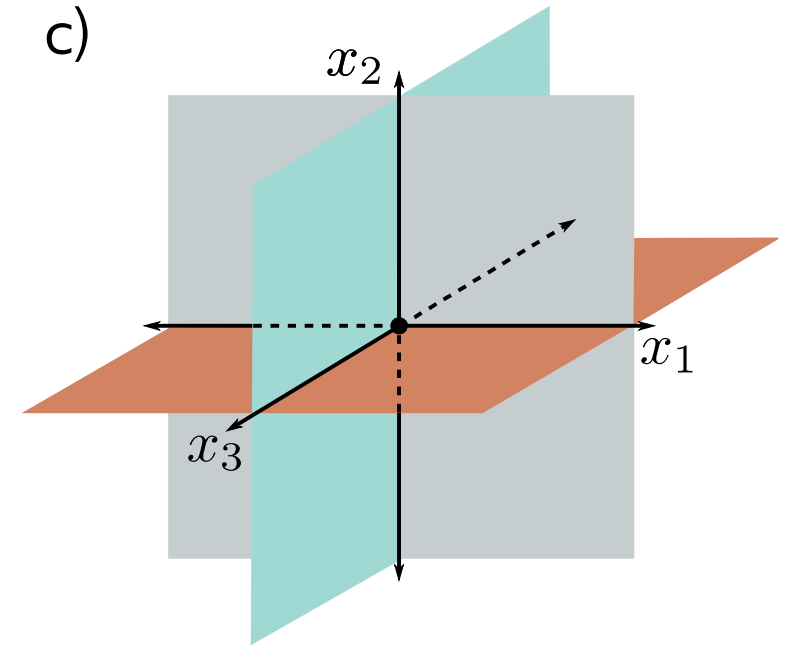
1D input with 1 hidden  
unit creates two regions  
(one joint)

b)



2D input with 2 hidden  
units creates four regions  
(two lines)

c)

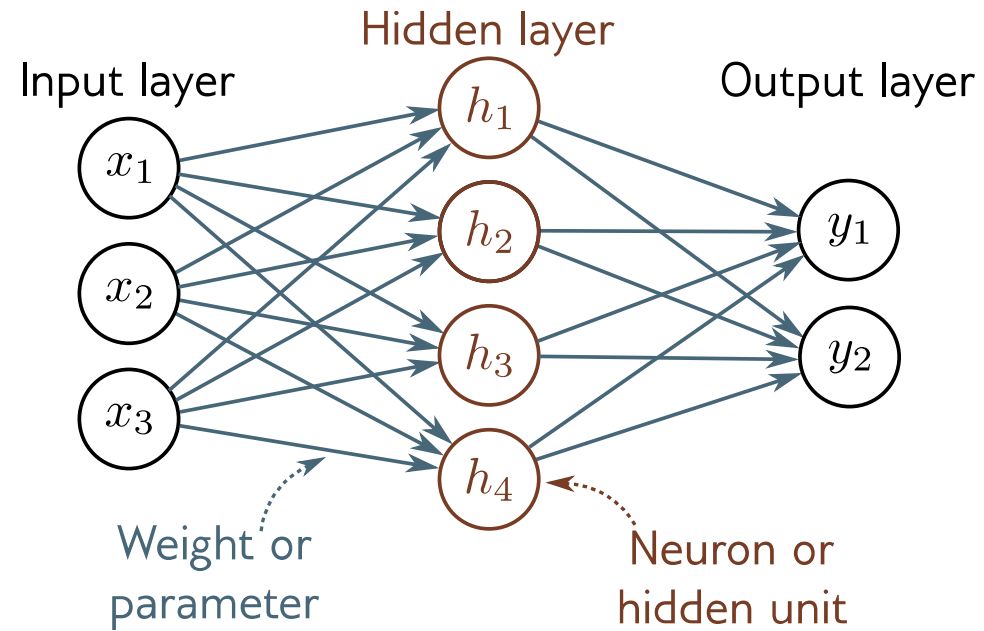


3D input with 3 hidden  
units creates eight regions  
(three planes)

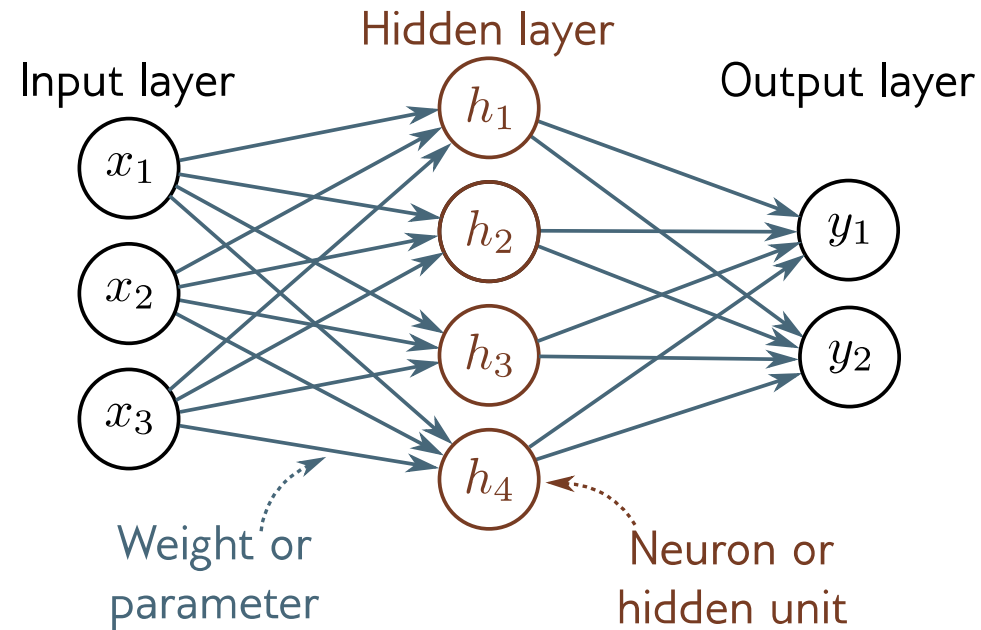
# Shallow neural networks

- Example network, 1 input, 1 output
- Universal approximation theorem
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- More than one input
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# Nomenclature



# Nomenclature



- Y-offsets = **biases**
- Slopes = **weights**
- Everything in one layer connected to everything in the next = **fully connected network**
- No loops = **feedforward network**
- Values after ReLU (activation functions) = **activations**
- Values before ReLU = **pre-activations**
- One hidden layer = **shallow neural network**
- More than one hidden layer = **deep neural network**
- Number of hidden units  $\approx$  **capacity**

# Other activation functions

